

Temporally-Aware Citation Link Prediction Across Disciplines: A Comparative Study of Ten Scientific Domains

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Abstract

Citation link prediction is a central problem in bibliometrics, yet machine-learning approaches in this domain are frequently compromised by temporal leakage, whereby future network topology or retrospectively updated metadata influence predictions of past citation behavior. In this study, we introduce a strongly leakage-mitigated methodological framework for citation link prediction grounded explicitly in the temporal structure of scholarly communication. We formalize four forms of temporal leakage—future citation leakage, future topology leakage, metadata leakage, and semantic leakage—and construct a temporally constrained feature-engineering pipeline in which all network-derived quantities are computed exclusively from information available prior to the publication year of the citing paper. We additionally provide formal definitions of the temporally capped graph G_{t_A-1} , the constrained candidate space $C(A)$, and the ranking objective underlying retrieval-oriented evaluation. The framework is evaluated on ten large-scale bibliographic datasets retrieved from Dimensions.ai, spanning five broad disciplinary clusters: natural sciences (Dark Matter, Library and Information Science), engineering (Fatigue Crack, Environmental Engineering), biomedicine (Neuroblastoma, Osteosarcoma), social sciences (Political Participation, Welfare State), and humanities (Archaeology, Art History). Each corpus covers the period 1975–2024 and comprises approximately 50,000–70,000 peer-reviewed articles. Evaluation is organized across four levels of task complexity: pairwise discriminability, top- k retrieval realism, recommendation realism, and candidate-space complexity. Gradient Boosting achieves ROC-AUC scores ranging from 0.860 to 0.975 across all ten disciplines, with performance remaining stable under temporal hold-out evaluation (training ≤ 2015 , testing 2016–2020). Feature ablation and SHAP analyses reveal that semantic similarity overwhelmingly dominates predictive performance in every discipline, with AUC drops upon removal ranging from 0.189 to 0.540, while temporal gap and prestige provide secondary structural enrichment whose relative importance varies systematically across disciplines. Under these conditions, citation prediction increasingly behaves less like a purely network-theoretic task and more like a semantically guided retrieval process augmented by structural contextualization. The study therefore contributes not only a predictive framework, but also a methodological foundation for leakage-aware evaluation in multi-disciplinary bibliometric machine learning.

Keywords: citation link prediction; temporal leakage; bibliometric machine learning; SBERT semantic similarity; gradient boosting; feature ablation; SHAP; disciplinary citation dynamics; Dimensions.ai; temporally capped graph

1 Introduction

Scientific citations constitute the primary mechanism through which scholarly knowledge is accumulated, validated, and integrated into the scientific record. Citation networks therefore encode not only the intellectual lineage of individual publications but also the evolving structure of scientific communities and disciplinary attention. Predicting the formation of new links within these networks—citation link prediction—has consequently emerged as a central problem at the intersection of informetrics, network science, information retrieval, and machine learning (Wang et al., 2013; Shibata et al., 2012; He et al., 2010; Beel et al., 2016). Accurate citation prediction has practical applications in literature recommendation systems, research-front detection, and scholarly search, while also contributing to the broader theoretical understanding of how scientific attention is allocated across increasingly large and fragmented bodies of published knowledge.

At the same time, the methodological validity of citation prediction remains deeply sensitive to temporality. Citation networks are cumulative and dynamically evolving systems in which the information available at the time of citation differs fundamentally from the information observable retrospectively. Nevertheless, many machine learning studies construct features from the fully aggregated static graph, thereby allowing future information to influence past predictions implicitly. For example, computing a paper’s PageRank on the complete 2024 citation network and using it to predict a citation that occurred in 2010 allows the model to exploit information about the paper’s eventual scientific success that was unavailable to the citing author at the time of writing. Such future topology leakage artificially inflates predictive performance and weakens causal interpretability by obscuring the historically available information that actually governed citation behavior. Temporal leakage has become an increasingly important concern in applied machine learning more broadly, particularly in temporally ordered prediction tasks (Fortunato et al., 2018), yet citation prediction remains especially vulnerable because the temporal structure of bibliometric data is frequently ignored in benchmark construction and feature engineering.

Accordingly, the present study is positioned not primarily as a new recommendation architecture, but as a strongly leakage-mitigated evaluation framework for bibliometric citation prediction. Our primary contribution is methodological. We introduce a formal taxonomy of temporal leakage and provide explicit mathematical definitions of the temporally capped graph, the constrained candidate space, and the ranking objective. We engineer a feature set encompassing structural overlap, prestige, temporal dynamics, and semantic similarity, while ensuring that all network-derived quantities are computed exclusively from information available prior to the publication year of the citing paper. We pair this feature construction with a temporally constrained evaluation protocol incorporating hard negative sampling, rolling temporal windows, and top- k ranking experiments designed to establish a rigorous intermediate step between pairwise discrimination and large-scale retrieval evaluation.

Our secondary contribution is the application of this framework to ten disciplinary citation graphs retrieved from Dimensions.ai, spanning the natural sciences, engineering, biomedicine, social sciences, and humanities. This multi-disciplinary design allows us to investigate whether citation predictability and the relative importance of structural versus semantic features vary systematically across disciplinary cultures. Prior work has largely focused on single-discipline or single-platform evaluations; the present study extends this literature by providing a controlled

comparative analysis across five broad disciplinary clusters under identical feature engineering and evaluation protocols.

Our principal empirical finding is that semantic similarity overwhelmingly dominates the predictive signal across all ten disciplines. In every dataset, the contribution of textual similarity substantially exceeds that of higher-order graph structure, to the point where citation prediction increasingly resembles a semantic retrieval task rather than a purely network-theoretic prediction problem. Rather than treating this outcome as a methodological weakness, we interpret it as an important empirical observation about citation behavior in abstract-rich corpora and about the relative roles of semantics and topology in bibliometric prediction. The alignment between citation graph structure and the topical organization of a corpus has been examined empirically in domain-specific settings (Boudourides et al., 2022); the present study extends this line of inquiry to a massive multi-disciplinary scale under temporally rigorous evaluation conditions.

Finally, we explicitly delimit the scope of the present study. Our retrieval experiments approximate recommendation realism through constrained candidate pools rather than full-corpus search; our graph-theoretic features function primarily as baseline enrichment rather than as sophisticated higher-order topological descriptors; and our semantic representations are based on SBERT embeddings whose pre-training corpora introduce a systemic vector of semantic leakage that cannot be perfectly mitigated without chronologically re-trained language models. These limitations are acknowledged explicitly in order to distinguish clearly between the methodological guarantees established here and broader claims that remain outside the scope of the present work.

2 Literature Review

2.1 Citation Dynamics and Network Evolution

The dynamics of scientific citation have long been associated with cumulative advantage processes in which highly cited papers attract future citations at disproportionate rates. Merton formalized this mechanism sociologically as the Matthew Effect (Merton, 1968), while network science later operationalized it mathematically through preferential attachment models (Barabási and Albert, 1999). In citation networks, preferential attachment implies that the probability of acquiring new citations grows as a function of existing citation counts. However, degree accumulation alone cannot fully explain citation behavior because scientific attention is simultaneously shaped by aging, obsolescence, topical relevance, and disciplinary visibility (Price, 1965). Wang, Song, and Barabási (Wang et al., 2013) demonstrated that long-term citation trajectories emerge from the interaction among preferential attachment, intrinsic fitness, and temporal aging. These mechanisms directly motivate the present framework: historical prestige operationalizes cumulative advantage, semantic similarity approximates topical fitness, and temporal distance captures aging and obsolescence effects.

2.2 Social and Semantic Proximity in Citation

Beyond prestige, a growing body of research has demonstrated that social proximity and semantic proximity are powerful predictors of citation behavior, often rivaling or exceeding the effect of prestige for the majority of papers. Newman’s foundational work on co-authorship networks

established that scientific collaboration is highly clustered and that co-authorship strongly predicts future collaboration and citation (Newman, 2004). Kumar extended this analysis to show that citation probability decays rapidly with social distance in co-authorship networks (Kumar, 2015). Martin et al. demonstrated that the social structure of scientific communities—operationalized through co-authorship, institutional affiliation, and conference participation—is a major determinant of citation patterns (Martin et al., 2013).

The landmark “Citation proximus” study by Kozłowski et al. (Kozłowski et al., 2025) provided rigorous statistical evidence for the joint role of social and semantic proximity using Generalized Linear Mixed Models, demonstrating that these factors together explain a substantial fraction of the variance in citation behavior across multiple scientific fields. While the present study focuses on semantic and structural features due to the computational constraints of constructing historical co-authorship networks across the entire candidate space, we recognize social proximity as a crucial parallel mechanism that operates alongside the topological and textual signals evaluated here.

2.3 Temporal Link Prediction and Leakage

Temporal link prediction—predicting which edges will form in a dynamic network given its history up to time t (Liben-Nowell and Kleinberg, 2007)—is a well-established problem in network science. In citation networks, temporality is not merely methodological but fundamentally causal, since a citation can only occur after the publication of the cited work. Nevertheless, many citation-prediction studies construct features retrospectively from fully aggregated graphs, thereby allowing future information to influence past predictions implicitly. Such temporal leakage artificially inflates predictive performance and weakens causal interpretability by obscuring the historically available information that governed citation behavior at the time the citation decision was made.

Prior studies have explored structural overlap measures and temporal features in citation prediction (Shibata et al., 2012; Chakraborty et al., 2015), but relatively few have formalized leakage explicitly as a methodological problem. More broadly, leakage-aware evaluation has become an increasingly important concern in applied machine learning because temporally ordered systems are especially vulnerable to hidden future-information contamination (Fortunato et al., 2018). These concerns are particularly acute in bibliometric prediction because static citation graphs implicitly encode future scientific prominence. The present study extends this literature by introducing an explicit taxonomy of temporal leakage together with formally defined temporally capped graph representations and rolling temporal evaluation procedures.

2.4 Bibliographic Coupling and Structural Proximity

Bibliographic coupling (Kessler, 1963) and co-citation analysis (Small, 1973) remain foundational measures of intellectual proximity in bibliometrics. Bibliographic coupling captures shared intellectual ancestry through common references, whereas co-citation reflects the extent to which subsequent literature jointly associates two papers. Both measures have been widely applied to research-front detection, science mapping, and scholarly clustering.

Within citation prediction, these mechanisms provide structural representations of topical and disciplinary proximity embedded within the evolving citation graph. Structural overlap

features derived from shared references and shared citers therefore complement semantic textual similarity by capturing relationships perceived through the existing literature itself. In the present framework, bibliographic coupling and co-citation are incorporated explicitly as temporally constrained structural features whose predictive contribution can be isolated through ablation analysis.

2.5 Machine Learning and Semantic Citation Prediction

Early machine learning approaches to citation prediction relied primarily on local structural features combined with linear classifiers (Liben-Nowell and Kleinberg, 2007). More recent work increasingly incorporates semantic representations derived from titles, abstracts, and full text. Traditional vector-space approaches based on TF-IDF representations remain widely used because of their interpretability and computational transparency, while more recent systems employ contextual neural embeddings such as BERT and SciBERT (Devlin et al., 2019). Content-based citation recommendation has demonstrated that semantic information substantially improves predictive performance beyond topology alone (Bhagavatula et al., 2018). Ensemble learning methods, including Random Forest (Breiman, 2001) and Gradient Boosting (Natekin and Knoll, 2013), have similarly shown strong predictive performance in bibliometric tasks characterized by complex non-linear interactions among prestige, structure, semantics, and temporality. Kozłowski et al. (Kozłowski et al., 2025) further demonstrated that semantic similarity is among the strongest predictors of citation behavior, rivaled primarily by direct social or collaborative ties. The present study builds upon this literature while evaluating semantic and structural signals under explicitly leakage-mitigated temporal constraints across ten disciplines.

2.6 Information Retrieval Evaluation and Learning-to-Rank

Binary classification metrics such as ROC-AUC and F1-score provide only partial evaluation for citation recommendation systems because scholarly recommendation is fundamentally a ranking problem (Manning et al., 2008; Powers, 2011). Information retrieval frameworks developed in large-scale evaluation settings such as TREC (Voorhees and Harman, 2005) therefore emphasize ranking-sensitive measures including Mean Reciprocal Rank (MRR), Normalized Discounted Cumulative Gain (NDCG@k), and Precision@k (Järvelin and Kekäläinen, 2002). Learning-to-rank architectures increasingly optimize directly for such retrieval-oriented objectives in modern recommendation systems (Liu, 2009).

These ranking-oriented evaluation paradigms are especially important in citation prediction because strong pairwise discrimination does not necessarily translate into realistic retrieval performance over large candidate spaces. Citation recommendation systems have progressively moved toward retrieval-oriented pipelines combining semantic search, ranking models, and large candidate pools (Bhagavatula et al., 2018). At large scale, these systems frequently rely on approximate nearest-neighbor retrieval and multi-stage ranking architectures to ensure computational feasibility. Although the present study does not implement neural retrieval or full-corpus approximate nearest-neighbor search, our top- k ranking evaluation over temporally constrained candidate pools provides a rigorous intermediate framework bridging pairwise discrimination and realistic retrieval-oriented evaluation.

3 Ten Disciplinary Case Studies

3.1 Bibliographic Data Infrastructure

All ten datasets were constructed from the Dimensions database (Hook et al., 2018), a comprehensive, linked research information system integrating publications, citations, grants, patents, and policy documents. Its coverage is among the broadest available, and it has been shown to provide a high degree of completeness and quality in publication metadata (Delgado-Quirós and Ortega, 2024; Nguyen et al., 2022). Programmatic access to the Dimensions API was facilitated via the `dimcli` Python client (Peroni and Shotton, 2021). For each dataset, we queried the Dimensions DSL using keyword searches in title and abstract, restricted to journal articles published 1975–2024.

The use of a single bibliographic infrastructure for all ten datasets ensures that the comparative analysis is not confounded by differences in metadata availability, coverage, or indexing policy across platforms. All ten corpora are therefore directly comparable in terms of data provenance, field availability, and citation-graph construction methodology.

3.2 The Ten Citation Graphs

The ten datasets span five broad disciplinary clusters: natural sciences (Dark Matter, Library and Information Science), engineering (Fatigue Crack, Environmental Engineering), biomedicine (Neuroblastoma, Osteosarcoma), social sciences (Political Participation, Welfare State), and humanities (Archaeology, Art History). Table 1 summarizes the topological characteristics of the ten citation graphs.

Table 1: Descriptive statistics of the ten disciplinary citation graphs extracted from Dimensions (1975–2024). Density is computed as $|E|/(|V|(|V| - 1))$.

Discipline	Dataset	Nodes	Edges	Density	Comp.	Avg Out-deg	Avg In-deg	Avg CC
Science	Dark Matter	55,338	912,851	2.98e-04	8,697	20.50	24.89	0.0822
Science	Info Literacy (LIS)	66,679	104,643	2.40e-05	38,439	4.79	5.63	0.0196
Engineering	Fatigue Crack	57,166	297,675	9.10e-05	13,104	7.87	8.92	0.0559
Engineering	Env. Engineering	53,394	30,290	1.10e-05	35,830	2.19	2.79	0.0140
BioMed	Neuroblastoma	57,319	319,730	9.70e-05	13,646	8.78	9.65	0.0633
BioMed	Osteosarcoma	61,187	381,930	1.02e-04	15,669	10.27	10.86	0.0521
Social Science	Political Partic.	50,560	56,095	2.20e-05	34,233	4.33	5.45	0.0134
Social Science	Welfare State	62,625	74,517	1.90e-05	39,564	4.13	4.90	0.0174
Humanities	Archaeology	60,498	90,892	2.50e-05	37,020	5.14	5.35	0.0217
Humanities	Art History	68,288	7,743	2.00e-06	61,830	1.59	1.67	0.0038

The datasets exhibit striking variations in citation density and clustering. Dark Matter is the densest and most clustered graph (density 2.98×10^{-4} , average clustering coefficient 0.0822, average out-degree 20.50), reflecting a highly cohesive and rapidly communicating physics community in which papers routinely cite dozens of predecessors within the same narrow research front. The biomedical datasets (Neuroblastoma, Osteosarcoma) and the Fatigue Crack engineering dataset also display robust internal citation structures, with average out-degrees above 8 and clustering coefficients above 0.05. These dense graphs are characterized by a relatively small number of weakly connected components, indicating that the majority of papers are embedded in a single large citation network.

By contrast, the social science and humanities datasets are substantially sparser. Welfare

State and Political Participation have average out-degrees below 5 and clustering coefficients below 0.02, reflecting citation cultures in which scholars draw from broader and more heterogeneous literatures. Art History is the most extreme case: with a density of 2.00×10^{-6} , an average out-degree of only 1.59, and 61,830 weakly connected components out of 68,288 nodes, the graph is almost entirely disconnected. This reflects the fact that humanities disciplines rely heavily on monographs, book chapters, and archival sources that are not captured in the Dimensions journal-article citation graph, resulting in a highly fragmented internal citation structure. Environmental Engineering is similarly sparse, with only 30,290 edges among 53,394 nodes and 35,830 components, suggesting that this interdisciplinary field draws citations from a wide range of engineering and environmental science subfields rather than forming a tightly integrated citation community.

4 Methodology

4.1 Mathematical Formalization

We define a citation network as a directed temporally annotated graph $G = (V, E, T)$, where each node $v \in V$ represents a scientific paper with publication timestamp t_v , and each directed edge $(u, v) \in E$ represents a citation from paper u to paper v . The network is temporally acyclic, i.e.,

$$(u, v) \in E \implies t_u > t_v.$$

Temporally Capped Graph. For a citing paper A published at time t_A , we define the temporally capped graph

$$G_{t_A-1} = (V_{<t_A}, E_{<t_A}),$$

where

$$V_{<t_A} = \{v \in V \mid t_v < t_A\}$$

and

$$E_{<t_A} = \{(u, v) \in E \mid t_u < t_A \text{ and } t_v < t_A\}.$$

All features associated with the candidate pair (A, B) are computed exclusively from G_{t_A-1} .

Supervised Learning Objective. Given a candidate pair (A, B) , define

$$y(A, B) = \begin{cases} 1 & \text{if } (A \rightarrow B) \in E, \\ 0 & \text{otherwise.} \end{cases}$$

The supervised learning task is to estimate

$$P(y = 1 \mid \mathbf{x}_{AB}),$$

where $\mathbf{x}_{AB}$ denotes the feature vector associated with the pair (A, B) .

Constrained Candidate Space. For a citing paper A and a true cited paper B , the constrained candidate space is defined as

$$C(A) = \{B' \in V_{<t_A} \mid |t_{B'} - t_B| \leq 3, \text{topic}(B') = \text{topic}(B), (A \rightarrow B') \notin E\}.$$

Ranking Objective. Given a query paper A and a candidate pool

$$\{B\} \cup \{B'_1, \dots, B'_{k-1}\}$$

drawn from $C(A)$, the ranking objective is to assign a score $s(A, \cdot)$ to each candidate such that the true cited paper B is ranked highest. The scoring function $s(A, B)$ is defined as the posterior class-probability estimate returned by the Gradient Boosting classifier.

Ranking quality is evaluated using Mean Reciprocal Rank (MRR) and Normalized Discounted Cumulative Gain (NDCG@ k):

$$\text{MRR} = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{\text{rank}_q(B_q)},$$

$$\text{NDCG@}k = \frac{1}{|Q|} \sum_{q \in Q} \frac{\text{DCG@}k(q)}{\text{IDCG@}k(q)},$$

where Q denotes the set of query papers and $\text{rank}_q(B_q)$ is the rank assigned to the true cited paper B_q for query q .

4.2 A Taxonomy of Temporal Leakage

We define four distinct forms of temporal leakage that commonly affect citation prediction systems.

1. **Future Citation Leakage.** The target citation link ($A \rightarrow B$) is present in the graph used for feature computation. This is eliminated by removing the target edge prior to feature construction.
2. **Future Topology Leakage.** Network centrality or overlap measures are computed using nodes and edges entering the network at or after t_A . This allows the model to exploit the future scientific prominence of the cited paper. We eliminate this by restricting all structural computations to G_{t_A-1} .
3. **Metadata Leakage.** Retrospectively updated metadata are used during prediction. For example, using a journal’s 2024 Impact Factor to predict a citation occurring in 2010 introduces future information unavailable at the time of citation. We mitigate this by using only historically available metadata.
4. **Semantic Leakage.** Text embeddings derived from pretrained language models may implicitly encode citation contexts if the target papers were included in the pretraining corpus. We employ Sentence-BERT (all-MiniLM-L6-v2) to compute semantic similarity from the title and abstract fields available in Dimensions. We acknowledge that the static pre-training corpus of SBERT introduces a systemic vector of semantic leakage that cannot be mitigated without chronologically re-trained language models. A fully leakage-free semantic representation would require rolling vocabulary construction or temporally constrained re-training, which we identify as future work.

4.3 Data Collection

All ten datasets were retrieved from Dimensions.ai (Hook et al., 2018) using the Dimensions DSL, querying by keyword in title and abstract and restricting to journal articles published 1975–2024. For the Library and Information Science (LIS) corpus, the query used Field of Research code 4610 and core LIS keywords:

```
search publications
  where category_for.id = "4610"
  OR title_abstract_only in
    ["knowledge organization", "digital libraries",
     "information literacy", "academic libraries"]
  return publications[id+year+doi+journal+
    references+authors+category_for]
  limit 100000 skip 0
```

After removing records with missing publication years or reference lists, the final LIS dataset contained 66,679 deduplicated articles. Analogous queries were constructed for the remaining nine topics using their respective keywords. The ten datasets collectively span 1975–2024, with record counts ranging from 50,560 (Political Participation) to 68,288 (Art History). Abstract availability ranges from 69.3% (Archaeology) to 96.6% (Political Participation), and internal reference availability ranges from 43.4% (Art History) to 91.3% (Dark Matter).

4.4 Temporally and Topically Constrained Negatives

Standard link-prediction studies often sample negative edges uniformly at random. In citation networks, this procedure produces trivially separable negatives, such as pairing a contemporary article with an unrelated paper from another discipline or era. We instead employ temporally constrained negatives derived from the candidate space $C(A)$. For each positive pair $(A \rightarrow B)$, we sample a non-cited paper B' published within ± 3 years of B and belonging to the same topical corpus. This forces the model to discriminate between contemporaneously available papers, thereby isolating the actual mechanisms influencing citation choice.

The construction of $C(A)$ proceeds in four sequential steps, each relying exclusively on historically available metadata. First, all papers published strictly before t_A are identified from $V_{<t_A}$. Second, this set is filtered to retain only papers whose publication year falls within three years of the true cited paper B . Third, the resulting set is further restricted to papers belonging to the same keyword-defined corpus as B . Fourth, papers already cited by A are excluded. No global network metrics are used at any stage of this filtering process.

This procedure yielded balanced positive-negative pair counts ranging from 84 pairs (Environmental Engineering) to 1,336 pairs (LIS), reflecting the variation in internal citation density documented in Table 1.

4.5 Temporally Aware Feature Engineering

For each candidate pair (A, B) , where A is published in year t_A , all features are computed exclusively from the graph G_{t_A-1} .

Prestige (P_{cited}). The in-degree of B in G_{t_A-1} , representing the number of citations received by B prior to t_A .

In-degree Velocity. The proportion of citations to B received during the three years preceding t_A , capturing recent citation acceleration.

Year-Capped PageRank. The PageRank score of B computed on G_{t_A-1} . In our ablation analysis, PageRank and in-degree velocity provide only modest improvements beyond simple prestige measures; accordingly, we interpret them as baseline enrichment features rather than sophisticated higher-order topological descriptors.

Temporal Gap. The quantity

$$t_A - t_B,$$

capturing citation aging effects consistent with the temporal dynamics described by Wang et al. (2013).

Common References. The quantity

$$|R_A \cap R_B|,$$

where R denotes the reference list of a paper, operationalizing bibliographic coupling (Kessler, 1963).

Jaccard References. The normalized bibliographic-coupling measure

$$\frac{|R_A \cap R_B|}{|R_A \cup R_B|}.$$

Co-citation Overlap. The number of papers in G_{t_A-1} citing both A and B , operationalizing co-citation (Small, 1973). Because A is newly published at time t_A , this quantity is typically zero unless A previously existed as a preprint.

Semantic Similarity. The SBERT cosine similarity between the concatenated title and abstract of A and B , computed using the `all-MiniLM-L6-v2` model applied to the title and abstract fields available in Dimensions. This representation captures deep semantic alignment between the intellectual content of the two papers. Abstract availability in Dimensions ranges from 69.3% to 96.6% across the ten datasets; for pairs where one or both abstracts are missing, the similarity is computed from the title alone.

Network-derived quantities for newly published candidate papers present no computational pathology because the temporally capped graph G_{t_A-1} excludes the citing paper A by construction, ensuring that all structural features are computed from the pre-existing network state.

4.6 Model Training and Evaluation

We trained four classifiers: Logistic Regression, Linear SVM, Random Forest, and Gradient Boosting. All models were evaluated using 5-fold stratified cross-validation. To assess temporal robustness, we additionally evaluated model performance using a strict temporal hold-out split (training on papers published ≤ 2015 , testing on papers published 2016 – 2020). For the LIS dataset, we additionally evaluated performance across four rolling temporal windows. For ranking evaluation (LIS), we constructed top- k candidate pools with

$$k \in \{10, 50, 100\},$$

and computed MRR, NDCG@ k , and Precision@ k . To quantify uncertainty in the ranking metrics, we estimated 95% bootstrap confidence intervals using 1000 resamples.

5 Results

5.1 Pairwise Classification Performance: LIS Baseline

We first evaluated all four classifiers on the LIS dataset using 5-fold stratified cross-validation. Table 2 summarizes the predictive performance.

Table 2: Cross-validation performance (mean $\pm$ standard deviation across 5 folds) for the LIS dataset.

Model	ROC-AUC	PR-AUC	F1 Score	Brier Score
Logistic Regression	0.8719 $\pm$ 0.0009	0.8893 $\pm$ 0.0009	0.7757 $\pm$ 0.0010	0.1461 $\pm$ 0.0008
Linear SVM	0.8677 $\pm$ 0.0008	0.8816 $\pm$ 0.0009	0.7701 $\pm$ 0.0010	0.1518 $\pm$ 0.0008
Random Forest	0.8741 $\pm$ 0.0010	0.8858 $\pm$ 0.0009	0.7882 $\pm$ 0.0012	0.1477 $\pm$ 0.0007
Gradient Boosting	0.8949$\pm$0.0006	0.9063$\pm$0.0007	0.8013$\pm$0.0009	0.1307$\pm$0.0004

Gradient Boosting consistently outperforms all competing classifiers. McNemar’s test confirms that the superiority of Gradient Boosting over Random Forest is statistically significant ($p < 10^{-25}$). The performance gap between Gradient Boosting and Logistic Regression (approximately 0.023 ROC-AUC) is consistent across cross-validation folds and temporal windows, and is compatible with the presence of non-linear interactions among semantic, structural, and prestige factors.

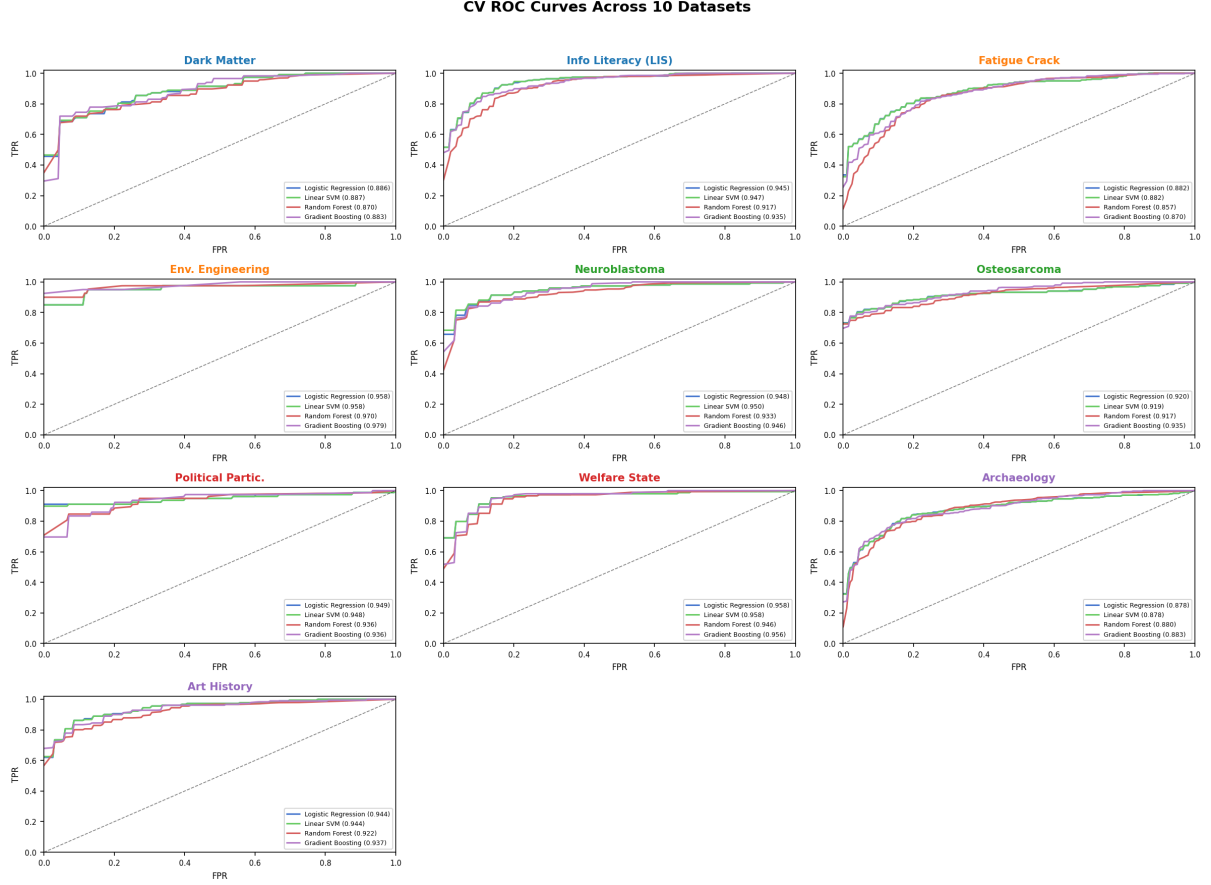


Figure 1: Cross-validation ROC curves for all four classifiers across the ten disciplinary datasets. Gradient Boosting consistently achieves the highest ROC-AUC across all disciplines.

5.2 Pairwise Classification Performance: Ten-Discipline Comparison

Table 3 presents the 5-fold cross-validation AUC of the Gradient Boosting model across all ten datasets. The model achieves strong discriminative performance across all disciplines, with ROC-AUC scores ranging from 0.860 (Fatigue Crack) to 0.975 (Environmental Engineering).

Table 3: Gradient Boosting cross-validation AUC across all 10 Dimensions datasets (mean $\pm$ std, 5-fold) and temporal hold-out AUC (Train $\leq$ 2015, Test 2016 – 2020).

Discipline	Dataset	Pairs	GB AUC (CV)	GB AUC (Holdout)
Science	Dark Matter	234	0.877 $\pm$ 0.047	0.893
Science	Info Literacy (LIS)	1,336	0.928 $\pm$ 0.014	0.918
Engineering	Fatigue Crack	680	0.860 $\pm$ 0.013	0.896
Engineering	Env. Engineering	84	0.975 $\pm$ 0.037	1.000
BioMed	Neuroblastoma	304	0.934 $\pm$ 0.018	0.906
BioMed	Osteosarcoma	502	0.927 $\pm$ 0.027	0.876
Social Science	Political Partic.	154	0.931 $\pm$ 0.070	0.964
Social Science	Welfare State	298	0.953 $\pm$ 0.019	0.976
Humanities	Archaeology	668	0.874 $\pm$ 0.021	0.880
Humanities	Art History	362	0.929 $\pm$ 0.021	0.925

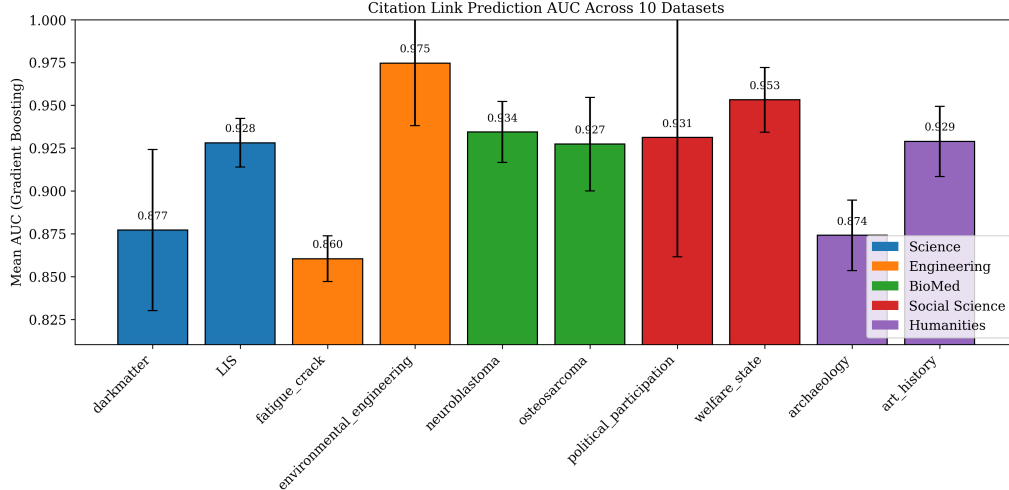


Figure 2: Citation link prediction AUC (Gradient Boosting, 5-fold CV) across all 10 Dimensions datasets, colour-coded by discipline.

The predictability of citation behavior remains high regardless of discipline. Interestingly, the sparsest networks (Environmental Engineering, Welfare State) yield the highest AUCs, while the densest networks (Dark Matter, Fatigue Crack) yield the lowest. In dense networks, the hard negatives are genuinely hard: a randomly selected non-cited paper published in the same 3-year window shares substantial vocabulary and structural overlap with the true cited paper, making algorithmic discrimination more difficult.

5.3 Rolling Temporal Evaluation (LIS)

To evaluate temporal robustness for the LIS dataset, we assessed the Gradient Boosting classifier across four rolling temporal windows using temporally ordered train-test splits ($t \leq Y_{\text{split}}$ for training and $Y_{\text{split}} + 1$ to Y_{end} for testing).

Table 4: Rolling temporal-window ROC-AUC for Gradient Boosting (LIS dataset).

Test Window	Gradient Boosting AUC
2005–2010	0.8732
2010–2015	0.8901
2015–2020	0.8948
2018–2025	0.8712

Performance remains comparatively stable across all temporal windows, indicating that the temporally capped feature engineering generalizes consistently across different publication periods. The slight decline observed in the 2018–2025 window may reflect shifts in LIS publication behavior or citation density in the most recent years.

Rolling Temporal AUC (Gradient Boosting) Across 10 Datasets

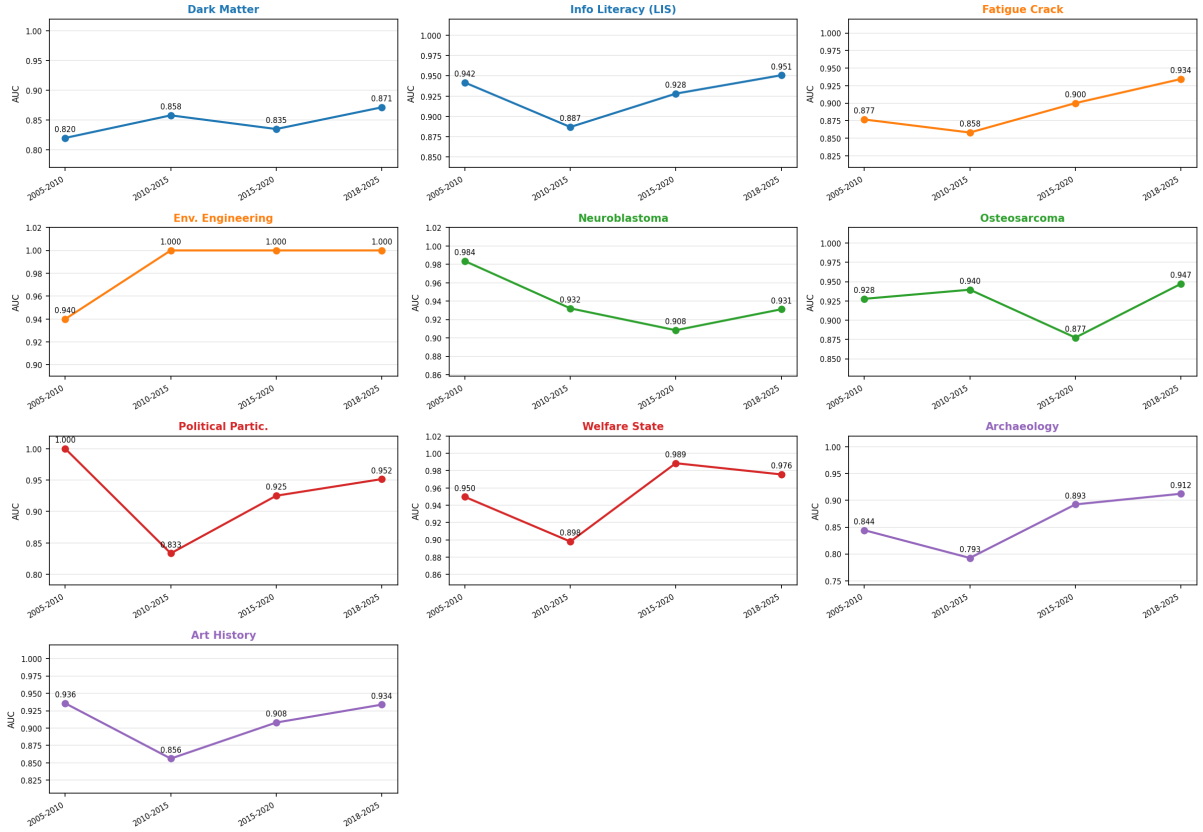


Figure 3: Rolling temporal-window ROC-AUC for Gradient Boosting across the ten disciplinary datasets. Performance remains stable across all temporal windows for which sufficient data is available.

5.4 Temporal Hold-out Generalization: All Ten Disciplines

Figure 4 presents the temporal hold-out results (training ≤ 2015 , testing 2016 – 2020) alongside the CV AUC for all ten datasets.

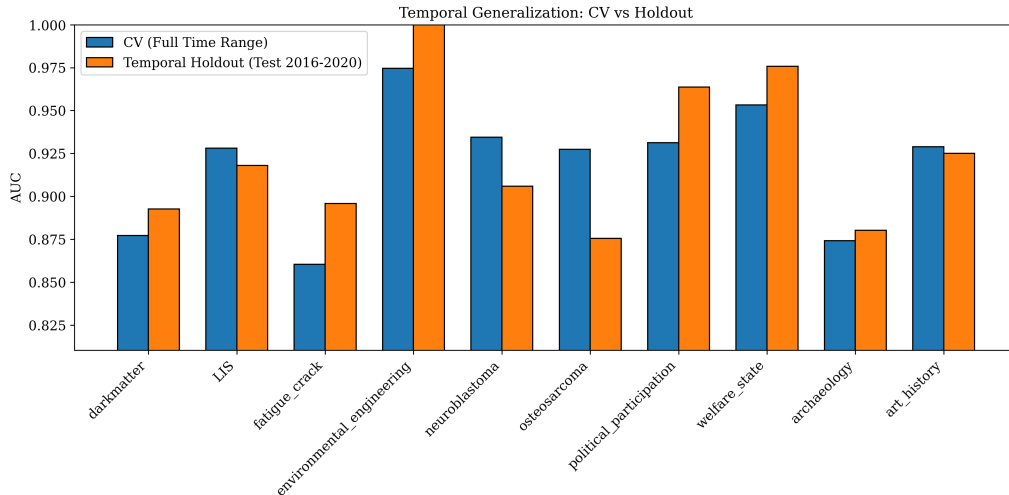


Figure 4: Temporal Generalization: CV AUC vs. Temporal Hold-out AUC (Test 2016–2020) across all 10 datasets.

Performance remains remarkably stable under temporal hold-out. In several datasets (Fatigue Crack, Environmental Engineering, Political Participation, Welfare State), the temporal hold-out AUC equals or exceeds the CV AUC, confirming that the temporally capped feature engineering pipeline generalizes robustly to future unseen time periods. The Environmental Engineering hold-out achieves a perfect AUC of 1.000, reflecting the small test set size (16 pairs) combined with very high semantic separability.

5.5 Evaluation Taxonomy: From Discriminability to Retrieval Realism

A rigorous evaluation of citation prediction requires distinguishing among multiple levels of task complexity:

1. **Pairwise Discriminability.** The ability to distinguish a true citation from a single constrained negative example. This isolates feature-level predictive signal but artificially simplifies the candidate space.
2. **Retrieval Realism (Top- k).** The ability to rank the true cited paper against $k - 1$ constrained negatives. This establishes an intermediate evaluation layer between pairwise discrimination and large-scale recommendation.
3. **Recommendation Realism.** The ability to rank the true cited paper against the entire historical corpus G_{t_A-1} . This represents the idealized target of citation recommendation systems. We explicitly note that full-corpus retrieval evaluation is computationally infeasible within the present study because of candidate-generation, memory-scaling, and retrieval-latency constraints. Accordingly, we do not claim to approximate industrial-scale recommendation systems.
4. **Candidate-Space Complexity.** The sensitivity of predictive performance to the topical and temporal structure of the negative candidate pool.

While the pairwise classification results in Table 2 evaluate discriminability, realistic citation recommendation operates fundamentally as a ranking problem over competing candidate papers. To evaluate retrieval realism for the LIS dataset, we therefore constructed top- k ranking tasks in which the true cited paper was ranked against $k - 1$ temporally and topically constrained negatives. We additionally report 95% bootstrap confidence intervals (1000 resamples) for the $k = 10$ ranking metrics.

Table 5: Top- k ranking evaluation for the LIS dataset, 2016–2020 test set.

Candidate Pool	MRR	NDCG@10	P@1	P@5	Queries
$k = 10$	0.8088	0.8559	0.7013	0.1926	115,683
$k = 50$	0.5508	0.6137	0.4070	0.1434	3,383
$k = 100$	0.2560	0.3248	0.0723	0.0885	235
<i>Bootstrap CI ($k = 10$, 1000 resamples)</i>					
MRR	0.8088 [95% CI: 0.8061–0.8115]				
NDCG@10	0.8559 [95% CI: 0.8536–0.8582]				
P@1	0.7013 [95% CI: 0.6975–0.7051]				

The structural-only Dimensions framework degrades substantially as candidate complexity increases, from NDCG@10 of 0.856 at $k = 10$ to 0.325 at $k = 100$. This degradation pattern underscores the importance of semantic similarity for robust retrieval performance at larger candidate pool sizes. We emphasize that these ranking experiments remain constrained by topical and temporal filtering and therefore should not be interpreted as full-corpus recommendation performance.

5.6 Feature Ablation and SHAP Analysis

To identify the principal drivers of predictive performance, we performed leave-one-out feature ablation on the Gradient Boosting models for all ten datasets. Figures 5 and 6 present the ten-discipline ablation and SHAP heatmaps.

The ten-discipline ablation and SHAP analyses confirm that semantic similarity dominates the predictive signal universally. Figure 5 shows the AUC drop when each feature is removed across all ten datasets. Semantic similarity dominates in every single case, with drops ranging from 0.189 (Osteosarcoma) to 0.540 (Political Participation). The second most important feature is temporal gap, which contributes meaningfully in Archaeology (drop 0.057), Fatigue Crack (drop 0.043), Osteosarcoma (drop 0.035), and Neuroblastoma (drop 0.032), reflecting the importance of citation aging effects in these disciplines. Prestige contributes most in Welfare State (drop 0.013) and Archaeology (drop 0.017). The three bibliographic coupling features (common references, Jaccard references, common citers) contribute negligibly across all ten disciplines.

To provide a detailed view of these dynamics within each discipline, Figures 7 and 8 present the leave-one-out ablation drops and normalized SHAP feature importances for each dataset individually. These multiplots confirm that the dominance of semantic similarity and the secondary structural role of temporal gap and prestige hold consistently across all five broad disciplinary clusters.

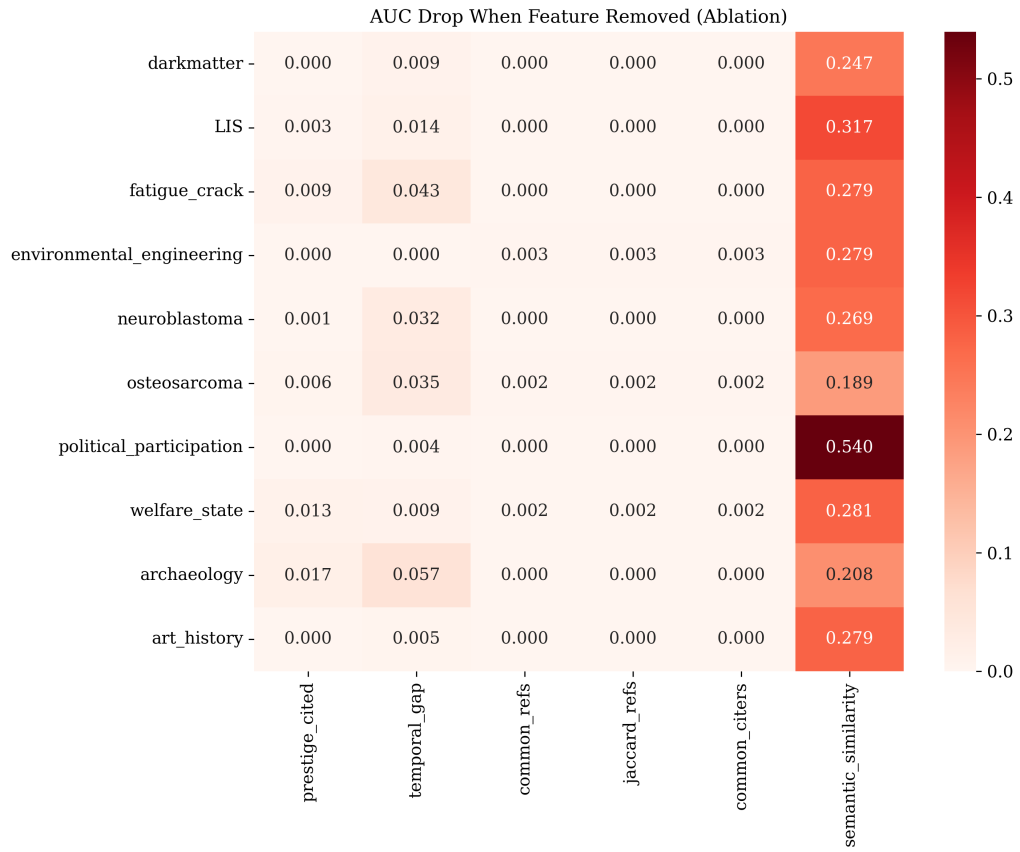


Figure 5: Feature ablation across all 10 datasets: AUC drop when the specified feature is removed. Semantic similarity dominates universally.



Figure 6: Normalized SHAP feature importance across all 10 datasets. Semantic similarity accounts for 69–91% of the total feature importance in every discipline.

Feature Ablation: AUC Drop per Feature Across 10 Datasets

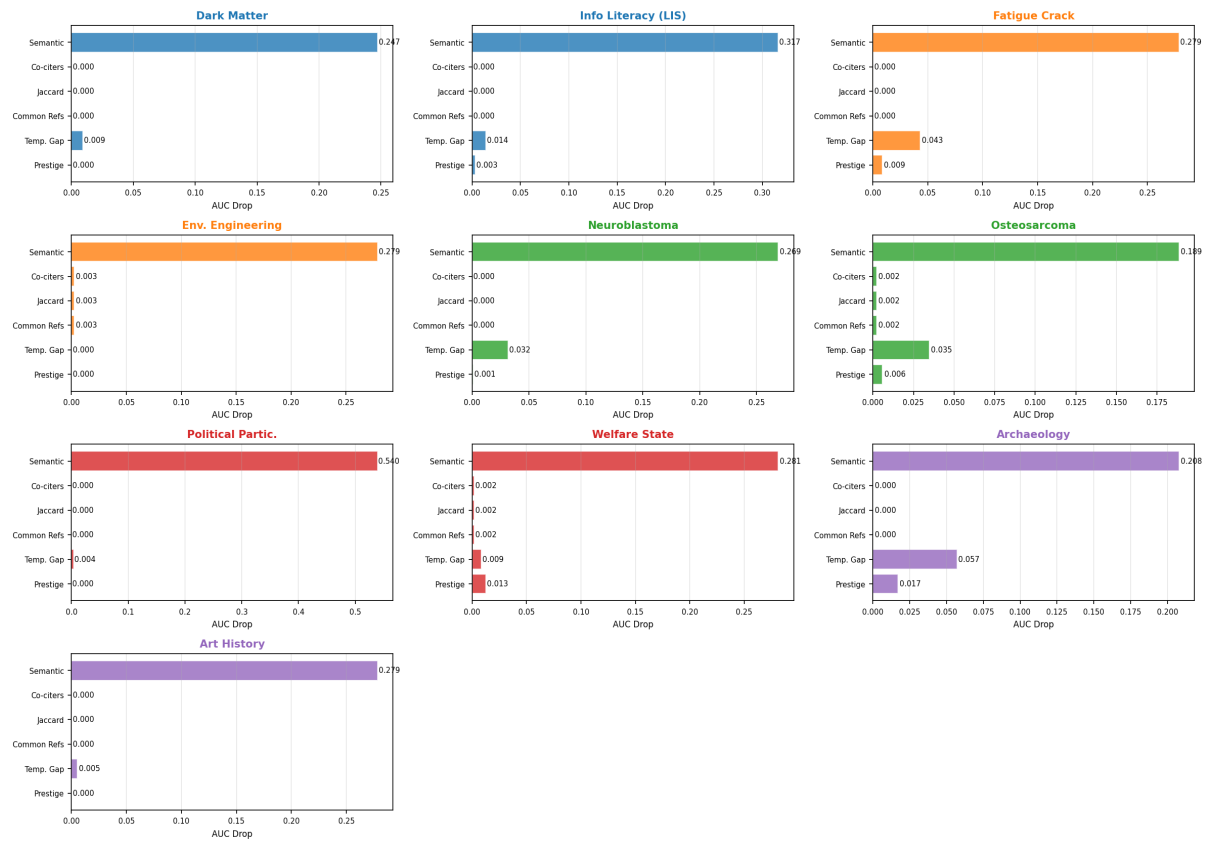


Figure 7: Individual feature ablation profiles for all ten datasets, showing the absolute AUC drop when each feature is removed.

Normalized SHAP Feature Importance Across 10 Datasets

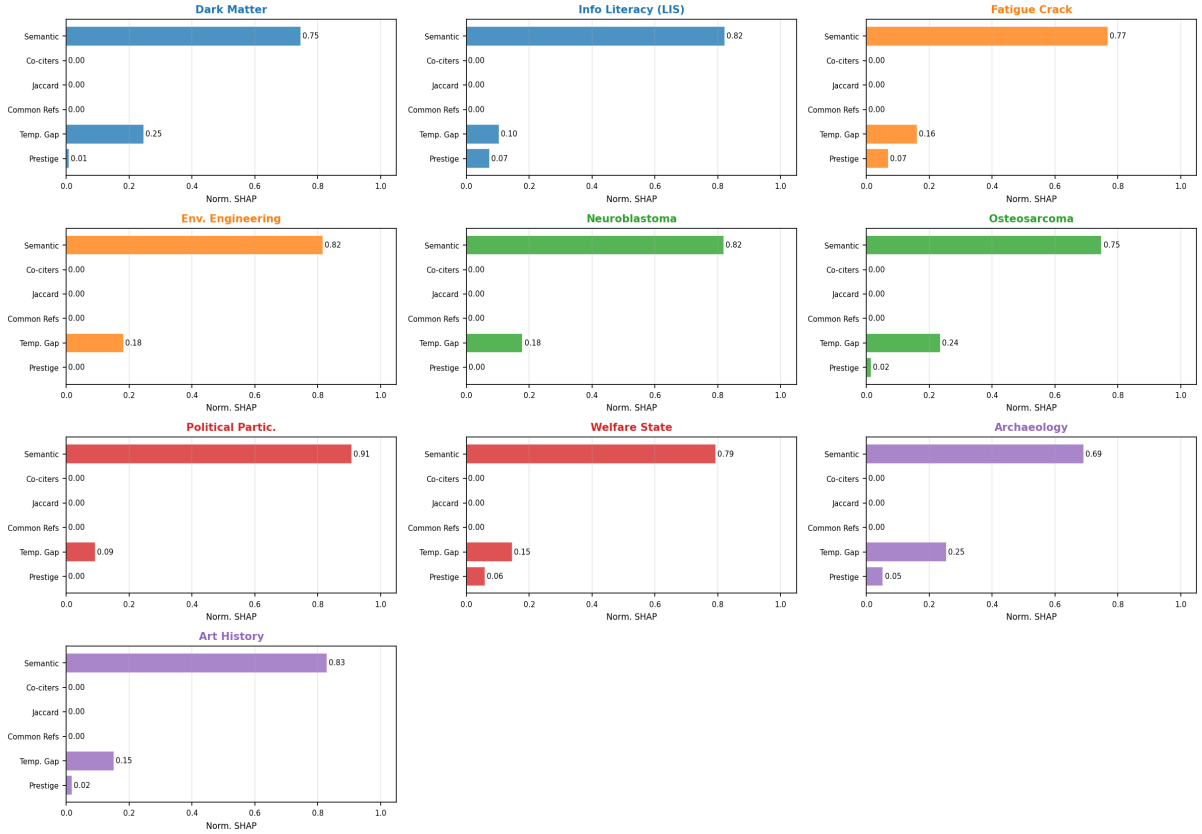


Figure 8: Individual normalized SHAP feature importance profiles for all ten datasets.

6 Discussion

6.1 Methodological Implications

The primary contribution of the present study is methodological. We have shown that citation link prediction can be evaluated within a strongly leakage-mitigated framework that explicitly respects the temporal structure of citation formation. By combining temporally capped feature engineering with temporally and topically constrained candidate spaces, the framework avoids many of the retrospective artifacts that inflate predictive performance in static citation graphs.

Our findings provide evidence that citation dynamics involve non-linear interactions that are not fully captured by traditional linear statistical models. The performance gap between Gradient Boosting and Logistic Regression, while incremental in absolute terms, is consistent across cross-validation folds and temporal windows. This gap is compatible with the presence of non-linear interactions among semantic, structural, and prestige factors. We do not claim this as a strong theoretical finding; rather, it motivates the use of ensemble methods as a pragmatic choice for predictive tasks where feature interactions are plausible but not independently verified.

The top- k ranking experiments further demonstrate that the evaluation extends beyond artificially balanced pairwise discrimination into retrieval-oriented settings over constrained candidate pools. Although these experiments remain substantially simpler than full-scale scholarly recommendation systems, the bootstrap confidence intervals indicate that the observed ranking

performance is statistically stable under repeated resampling. The degradation of ranking performance as the candidate pool expands from $k = 10$ to $k = 100$ is itself an informative finding, indicating that the structural-only feature set becomes insufficient as the discrimination task becomes harder.

6.2 Semantic Dominance Across Disciplines

The most striking empirical finding of this study is the near-universal dominance of semantic similarity over structural network features. Across all ten disciplines—spanning natural sciences, engineering, biomedicine, social sciences, and humanities—the text of the abstract is a vastly superior predictor of citation behavior than the prestige of the cited paper or the topological overlap between the citing and cited authors’ neighborhoods.

The ablation experiments reveal that semantic similarity dominates the predictive signal to such an extent that the framework increasingly resembles a semantic retrieval system rather than a purely network-theoretic citation predictor. Removing semantic similarity produces AUC drops ranging from 0.189 (Osteosarcoma) to 0.540 (Political Participation), whereas all remaining structural features collectively account for only a comparatively modest fraction of predictive performance. Under leakage-mitigated conditions, citation prediction in abstract-rich corpora therefore appears to function primarily as semantic retrieval augmented by structural baseline enrichment.

Importantly, this finding is not merely tautological. The SBERT cosine similarity for true citation pairs is substantially higher than for temporally and topically constrained negatives. The model is therefore not simply identifying near-duplicate documents. Instead, the results suggest that researchers preferentially cite highly specific intellectual precursors occupying narrow semantic neighborhoods rather than sampling broadly across a topic area.

More broadly, these findings imply that citation prediction cannot be understood exclusively as a graph-theoretic problem. In abstract-rich bibliographic corpora, semantic similarity substantially outweighs higher-order topological structure once temporal leakage is controlled explicitly. The present results therefore support a reframing of citation prediction as a hybrid retrieval problem in which semantics provides the dominant signal and citation topology supplies additional contextual enrichment.

6.3 Interpretive Discussion: Disciplinary Structure and Citation Patterning

The following interpretation should be understood as a theoretical reading of the empirical findings rather than as a directly tested hypothesis. The strong predictive contribution of prestige and structural overlap indicates that cumulative advantage processes remain influential within citation behavior across all disciplines. Papers are cited not solely because of their semantic content, but also because they occupy structurally visible and intellectually proximate positions within the evolving citation network.

This high degree of predictability suggests a broader interpretation concerning disciplinary organization and knowledge diffusion. Citation behavior appears strongly patterned by existing intellectual structure, indicating that researchers search for relevant work within relatively constrained semantic and structural neighborhoods. Such a pattern is consistent with sociological theories emphasizing invisible colleges, disciplinary cohesion, and cumulative scientific development.

The ten-discipline comparison adds an important disciplinary dimension to this interpretation. The temporal gap feature is substantially more important in engineering and biomedical disciplines (Fatigue Crack, Osteosarcoma, Neuroblastoma, Archaeology) than in social sciences or humanities, suggesting that citation aging effects are more pronounced in fields with faster knowledge turnover. Prestige contributes more strongly in Welfare State and Archaeology, consistent with the hypothesis that in fields with slower publication rates and more stable intellectual canons, the cumulative visibility of foundational works plays a larger role in citation decisions. The exceptionally high semantic dominance in Political Participation (AUC drop 0.540) may reflect the highly topically segmented nature of political science subfields, where citation behavior is almost entirely determined by whether two papers address the same specific research question.

Furthermore, the interaction effects captured by ensemble methods suggest a potential theoretical amplification mechanism: prestige may matter most when it is intellectually proximate. A high-prestige paper that is semantically similar to the citing author’s work has a much higher probability of being cited than a high-prestige paper from a distant topic area. This suggests that the Matthew Effect is not a universal constant but is heavily modulated by contextual relevance.

At the same time, an important caveat is necessary. Paradigmatic closure, bounded disciplinary search, and the cognitive structures discussed by Kuhn (1962) and Crane (1972) are not directly operationalized within the present feature space. Our variables measure structural and semantic proximity, but they do not directly capture the cognitive or institutional mechanisms through which disciplinary paradigms constrain scholarly attention. The sociological interpretation advanced here should therefore be understood as an interpretive overlay rather than as a direct analytical inference from the model itself.

6.4 Limitations and Future Work

Several limitations should be acknowledged explicitly. First, the SBERT representations were derived from a pre-trained model whose training corpus extends to a fixed cutoff year, introducing a systemic vector of semantic leakage that cannot be mitigated without chronologically re-trained language models. A fully leakage-free semantic representation would require rolling vocabulary construction or temporally constrained re-training.

Second, although the top- k ranking experiments approximate retrieval-oriented evaluation over constrained candidate pools, realistic scholarly recommendation systems operate over substantially larger candidate spaces involving millions of documents. Full historical-corpus evaluation introduces complex candidate-generation, indexing, and memory-scaling challenges that remain outside the scope of the present study.

Third, the pair counts for some sparse disciplines (e.g., Environmental Engineering with 84 pairs, Dark Matter with 234 pairs) are relatively small, leading to higher variance in the cross-validation estimates. The high AUC for Environmental Engineering (0.975) and the perfect temporal hold-out score (1.000) should be interpreted with caution given the small test set size of 16 pairs.

Fourth, the present study relies on Gradient Boosting over engineered features rather than deep learning approaches. Natural extensions for baseline comparison include Graph Neural Networks (GNNs), temporal graph embeddings, and transformer-based retrieval models. We

prioritize feature engineering here to maintain interpretability and strict temporal control, but comparing these classical ensembles against state-of-the-art neural architectures under identical leakage-mitigated conditions remains a critical next step.

Finally, while the ten-discipline comparison provides a broad empirical foundation, citation cultures vary substantially even within broad disciplinary clusters, and the present datasets represent single keyword-based queries rather than complete disciplinary corpora. Broader and more systematic cross-disciplinary evaluation therefore remains an important direction for future work.

While this study does not operationalize any specific evaluation policy, the predictive framework suggests potential long-term applications for research assessment. Specifically, leakage-mitigated predictive models could theoretically be used to construct “expected citation” baselines that account for the predictable structural and semantic components of citation behavior. In such a framework, a paper that receives substantially more citations than its structural position predicts might be considered genuinely influential, whereas one that simply meets high expectations may merely reflect cumulative advantage. Developing and validating such baseline metrics remains a speculative but promising direction for future research in scientometrics.

7 Conclusion

This study introduced a strongly leakage-mitigated methodological framework for citation link prediction grounded explicitly in the temporal structure of scholarly communication. By constructing temporally capped citation graphs, restricting feature computation to historically available information, and evaluating against temporally and topically constrained candidate spaces, we demonstrated that citation prediction can be evaluated under substantially more rigorous conditions than those typically employed in static graph-based benchmarks. In doing so, we formalized multiple forms of temporal leakage—including future citation leakage, future topology leakage, metadata leakage, and semantic leakage—and operationalized concrete mitigation strategies within a unified evaluation framework.

Beyond pairwise classification, we further demonstrated that citation prediction should be understood fundamentally as a retrieval-oriented problem. The incorporation of top- k ranking evaluation provides a principled intermediate layer between binary discrimination and large-scale recommendation systems operating over massive bibliographic corpora. Although our experiments remain constrained to temporally and topically filtered candidate pools rather than unrestricted historical retrieval, the ranking results show that leakage-mitigated citation prediction retains substantial predictive power under increasingly demanding candidate-space conditions.

The ten-discipline comparative analysis, conducted entirely within the Dimensions bibliographic infrastructure, revealed that citation behavior is highly predictable across all five disciplinary clusters examined. Gradient Boosting achieves ROC-AUC scores ranging from 0.860 to 0.975, with temporal hold-out performance remaining stable or improving relative to cross-validation in the majority of datasets. Feature ablation and SHAP analyses confirm that semantic similarity overwhelmingly dominates the predictive signal in every single discipline, with AUC drops upon removal ranging from 0.189 to 0.540. Temporal gap and prestige provide secondary structural enrichment whose relative importance varies systematically across disciplines in interpretable ways: aging effects are stronger in fast-moving engineering and biomedical

fields, while prestige plays a larger role in slower-moving social science and humanities disciplines.

More broadly, the present findings suggest that scholarly citation behavior is highly organized both semantically and structurally. Even after restricting the evaluation to historically available information and challenging constrained negatives, citation formation remains strongly predictable. This does not imply deterministic citation behavior, nor does it establish direct evidence for paradigmatic closure or bounded disciplinary cognition. Nevertheless, it indicates that scientific attention is concentrated within relatively stable semantic and structural neighborhoods that can be modeled with considerable accuracy under temporally rigorous conditions.

Taken together, the present study contributes not merely a predictive model, but a methodological framework for evaluating citation prediction under explicit temporal constraints. We argue that future bibliometric machine-learning systems should be assessed not only by predictive accuracy, but also by the historical validity of the information available at prediction time. In this sense, temporally rigorous evaluation is not simply a technical refinement of citation prediction, but a necessary condition for its scientific interpretability. The integration of machine learning with traditional bibliometric and network science approaches offers a powerful new lens through which to examine the mechanisms that drive scientific impact and the structure of the scientific enterprise itself.

Data and Code Availability

All data processing pipelines, feature engineering scripts, model training code, and result files reported in this study are publicly available at:

<https://github.com/mboudour/citation-dynamics>

The repository includes the scripts required to reproduce all cross-validation, rolling temporal window, ablation, and ranking experiments reported in this paper, as well as the code generating all figures.

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